DRUGS · EAR PREPARATIONS

Quinolone ear drops: ciprofloxacin and ofloxacin

Ciprofloxacin and ofloxacin ear drops are fluoroquinolones applied topically, alone or with a steroid. They cover Pseudomonas and Staphylococcus aureus, the two main organisms of otitis externa, and, unlike aminoglycoside drops, they have shown no ototoxicity even when they reach the middle ear. That makes them the first-line drops for a discharging ear with a perforation or a ventilation tube, and a good choice for otitis externa in general.

1 CHAPTER ONE At a glance

The drops on one page.

1.1 Summary

QUINOLONE EAR DROPS

	SUMMARY
Preparations	Ciprofloxacin 0.2% or 0.3% (single or with dexamethasone 0.1%); ofloxacin 0.3% (eye drop formulations often used in the ear)
How they work	Inhibit DNA gyrase and topoisomerase IV; topical concentrations far exceed the MIC of most organisms, including many strains "resistant" on systemic breakpoints
Spectrum	Pseudomonas aeruginosa, S. aureus (including many MRSA at topical concentrations), Proteus, H. influenzae; not fungi, limited against streptococci and anaerobes
ENT uses	Acute otitis externa; tympanostomy tube otorrhoea; chronic suppurative otitis media with discharge; discharging mastoid cavity; after ear surgery
Dose	Typically 4 drops (or one single-dose vial) twice daily for 7 days, varying by product; ofloxacin 10 drops once daily in adults (5 in children) in some licences
Children	Licensed in children with grommets (ciprofloxacin-dexamethasone from 6 months)

	SUMMARY
Pregnancy	Negligible systemic absorption; generally acceptable

2 CHAPTER TWO How they work

Mechanism and handling.

2.1 Why topical treatment wins

Ear drops deliver antibiotic at concentrations a thousand times higher than oral treatment achieves in the canal or middle ear, so they cure acute otitis externa and tube otorrhoea more often than oral antibiotics, without systemic side effects or resistance pressure on the gut flora. Systemic antibiotics are needed only when infection spreads beyond the canal (cellulitis), in the immunocompromised, or in necrotising otitis externa.

2.2 Getting the drops in

They work only if they reach the infection. **Aural toilet** (microsuction or dry mopping) first; a **wick** when the canal is swollen shut; lie with the ear up for 3 to 5 minutes after instilling, and **tragal pumping** to push drops through a tube or perforation.

3 CHAPTER THREE Using them in ENT

Indications and doses.

3.1 Acute otitis externa

Topical antibiotic with or without steroid for 7 days is the treatment; adding a steroid speeds pain relief. Continue a few days after symptoms settle; review if no

better in 48 to 72 hours.

3.2 Discharging ear with a perforation or a tube

Quinolone drops are first-line for **tympanostomy tube otorrhoea** (better than oral amoxicillin or watchful waiting) and for active **chronic suppurative otitis media**, because they are not ototoxic. Swab persistent discharge; consider fungal infection, cholesteatoma or a tumour if it does not settle.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- **Otomycosis**: prolonged antibacterial drops alter the canal flora; limit courses to 7 to 10 days and suspect fungi when discharge continues.
- Local irritation, itching, a bitter taste if drops pass through a perforation.
- Contact sensitivity is less common than with neomycin.

4.2 Cautions

Avoid in known quinolone hypersensitivity. They do not cover fungi.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

> **WHAT THE EXAMINER IS LOOKING FOR**
>
> - Why topical beats systemic for otitis externa and tube otorrhoea.

- Why quinolones are preferred when the drum is **not intact**.
- The role of aural toilet, wicks and tragal pumping.
- Otomycosis after prolonged use.

5.2 Common pitfalls

COMMON PITFALLS

- Prescribing drops into a canal full of debris.
- Oral antibiotics for uncomplicated otitis externa.
- Courses of several weeks that produce otomycosis.

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